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ORIGINAL ARTICLE

Teleconsultation in type 1 diabetes mellitus (TELEDIABE)

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Abstract

Aims The growing incidence of diabetes and the need to contain healthcare costs empower the necessity to identify new models of care. Telemedicine offers an acknowledged instrument to provide clinical health care at a distance, increasing patient compliance and the achievement of therapeutical goals. The objective was to test the feasibility and the efficacy in the improvement of the glycemic control of the teleconsultation for patients with type 1 diabetes mellitus.

Methods A randomized open-label, parallel arms, controlled trial was conducted in two diabetes centers in Italy. Participants affected by type 1 diabetes mellitus have been randomly (1:1) assigned to receive their visits as standard or a web-based care. Patients in the teleconsultation group can arrange their appointments on a Web site and can also have access to web educational courses or to nutritional and psychological counseling. The primary outcome was the assessment of glycemic control by HbA1c measurement after a 12-month follow-up.

Results Overall 74 participants were followed for 1 year. HbA1c changes were not statistically different within ($p = 0.56$ for standard care group; $p = 0.45$ for telemedicine group) and between ($p = 0.60$) groups when considering differences from baseline to the end of the study. Patients randomized to teleconsultation reported reduced severe hypoglycemic episodes ($p = 0.03$). In addition, they were largely satisfied with the activities, perceived a good improvement in the self-management of the diabetes, and reported to have a time saving and a cost reduction.

Conclusions In conclusion, TELEDIABE proposes a new system for the management of patients with type 1 diabetes mellitus.

Keywords Telemedicine · Skype · Type1 diabetes · HbA1c

Abbreviations

ICT Information and communications technology
DM Diabetes mellitus
CSII Continuous subcutaneous insulin infusion
MDI Multiple dose insulin

Introduction

Diabetes mellitus represents one of the major chronic diseases, and it is also a heavy healthcare financial burden, imposing a large economic load on individuals and families, national health systems, and countries. There is a large disparity in health spending on diabetes between regions and countries. Implementing inexpensive, easy-to-use interventions can reduce the huge economic costs of diabetes.

Research of new models in care organization demonstrates that advanced ICT (Information and Communications Technology) systems and services may have, among others, the potential to respond to the increasing burden of diabetes and the complexity of comorbidities and, doing so, to contribute to the sustainability of health and care systems.

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In this respect, Telemedicine offers a potential solution, especially with teleconsulting (a healthcare consultation carried out remotely with audiovisual telecommunications between doctor and patient), telenursing (remote therapeutic education), and telemonitoring (a system that monitors the condition of a patient in a different location). In particular, teleconsultation might provide a new approach to develop integrated care models that are more closely oriented to the needs of patients. There are some systematic reviews and meta-analysis indicating that the improvement of glycemic control in the group of patients followed by ICT solutions, compared to the standard examinations and checkups in the hospital, has led to an average reduction of about 0.4–0.5% glycated hemoglobin (HbA1c) [1–5]. These data were obtained in both adults and children. The benefit appears to be greater for those young patients with a history of poor glycemic self-control [3, 6–8], or with a lower socioeconomic level [9–11]. Tele-educational classes for patients to promote the learning of self-management of blood glucose monitoring, nutrition, and therapy have been also proposed [12–15]. The web holds the possibility to provide health information and support for many participants at relatively low cost and with a great freedom for patients to decide when and where to learn. There is evidence of a steady, though modest improvement in clinical parameters of glycaemia, even in older populations that are typically thought to be less accustomed to the use of the technology [16, 17]. Telemedicine could also facilitate the access to health services, especially for those families that cannot easily obtain absences from work or do not have a good mobility.

Among others, advanced ICT systems and services may have the potential to respond to the complexity of the insulin therapy management. The management of insulin therapy is often problematic due to the several variables that affect glucose homeostasis and hence exogenous insulin requirement; the most important are food, physical activity, stress, and illness. Therefore, patients under insulin treatment often suffer for severe glucose oscillations and hyperglycemic or hypoglycemic episodes. However, despite these premises, the management of diabetes is still anchored to traditional pathways. Although many experiences of web-based data sharing have been recently introduced (mainly glycemic profiles and laboratory results), the pillar of the management of patients with diabetes mellitus is still based on the visits in outpatient clinic.

The aim of our study is to propose teleconsultations as an innovative way of caring for people with type 1 diabetes mellitus under insulin therapy than the traditional outpatient clinic visits.

Methods

Study design

This is a randomized controlled study, open-label, parallel arms, involving subjects with type 1 diabetes. The two arms of the study were the control one, in which patients were submitted to standard visits in outpatient clinic, and the experimental one in which patients were submitted to teleconsultation. The assigned intervention was therefore a way of consultation: a standard visit in control group or a teleconsultation in the experimental group.

The teleconsultation with the diabetologist consisted in video communication that allows the maintenance of the doctor–patient relationship despite the physical distance. Before the teleconsultation, patients were invited to send via mail their glycemic profiles, the reports of HbA1c or of other biochemical analyses and of eventual medical examinations. Most of glucometers and continuous glucose sensors are supplied with a software for the web transmission of the data that facilitates the sharing of the information. During the visits, patients and physicians discussed the general clinical conditions, analyzed the data, reevaluated therapy, and planned proper follow-up. Apart from the teleconsultation, patients could access to tele-educational tools (online training courses previously uploaded to the project portal about nutrition, glycemic self-control, self-management of the therapy) as integration of what discussed during the visits. The patients could also request a teleconsultation with nutrition and psychological specialists.

The patients could look at the framework of the project in the Web site portal of the http://www.telediabete-fid.it/. Once registered and logged (the number of access is case sensitive), they could have an overview of the project, book the teleconsultation, access to the tele-educational tools, access to a chat open for the discussion between patients and physicians. The diabetologists or the patients could organize a supplementary clinical assessment in outpatient clinic in case patient needed a direct physical examination. The protocol has not been a constraint for the specialist in choosing the diagnostic and treatment path to follow.

The study was accepted for being supported by the Italian Foundation for Diabetes. The study was approved by our Ethical Committee (Board Approval Number *535-B-102014 by Comitato Etico Milano Area C) and then registered to ClinicalTrials.gov (ID: NCT02812498). The study started on January 2015 and was conducted until January 2017.

Primary objective of the study

The aim of the study has been the improvement of HbA1c at 12-month follow-up in patients with type 1 diabetes achieved by a service of tele web counseling compared to those patients followed by a standard visit in outpatient clinic. HbA1c is the main parameter representing the overall glycemic control and has been assessed on 3–4 months bases, but the value at 12 months is more representative of a sustained effect.

Secondary objectives of the study

The secondary objectives were:

1. The improvement of HbA1c at 4 and 8 month follow-up achieved by a service of tele web counseling compared to those patients followed by a standard visit in outpatient clinic.
2. A reduction in acute complications (severe hypoglycemic episodes requiring third assistance or access to the hospital, ketoacidosis)
3. The assessment of the patient satisfaction measured through a questionnaire that evaluated

- patient satisfactory assessment on their perception of advantages (diabetes management, comfort, and convenience) and limits (technical problems, poor interaction with physicians) of the service. A score of 1 (low) to 5 (High) has been used
- an estimate of the time and economic savings at 12-month follow-up in patients with type 1 diabetes achieved through a service of tele web counseling. Patients were asked to quantify the time and the economic savings. The eligible costs were the following: travel (fuel, bus or train tickets, parking), loss of working hours. The time considered to be eligible for the analyses was: the time requested for booking the appointment, for reaching the hospital, and for attending the visit. The patients were asked to estimate an average value for each visit.

Schema of the activities

Figure 1 describes the schema of the activities.

Visit 1: After the verification of the inclusion and of the exclusion criteria, the overview of the project was presented to the patients during the first visit. In case the patients agreed to the participation, the consent form to participate in the study was signed. Randomization to one of the treatment groups was done after the verification of the inclusion and of the exclusion criteria. The enrollment period was 12 months. The patients were randomized as follows:

Group A a clinical assessment of glycemic control through standard outpatient periodic evaluations.

Group B a clinical assessment of glycemic control through teleconsultation via web. Randomization was performed with the use of a computer-generated random sequence.

The patients were investigated for the presence of underlying diseases and any current or chronic intake of drugs. For each patient, it was analyzed the level of HbA1c by venipuncture already performed by diagnostic practices.

The visit 2 took place about 4 months from the previous visit. The visit 3 took place about 4 months from the previous visit and 8 months after the visit 1. The visit 4 took place 12 months from the visit 1. During each visit, glucose profiles and eventual clinical medical problems were discussed with the specialist. HbA1c was assessed before the visit. In the occasion of the visit at 12 months, the patients followed via web completed a questionnaire about a general evaluation of their consultations and of the project.

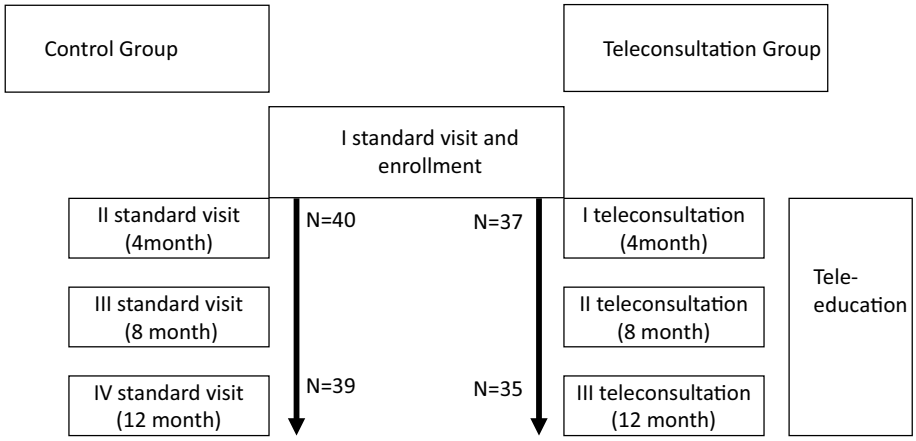

Fig. 1 Schema of the project

Study population and sample size

The following inclusion and exclusion criteria were defined. The patients with the following inclusion criteria were eligible for the study:

- Males and females aged 5–50 years old with type 1 diabetes in multi-insulin injection therapy or insulin pump.
- All patients having the above characteristics necessarily had to sign the informed consent for the inclusion in the study.
- Patients with Internet access.

The patients with the following exclusion criteria were not eligible for the study:

- Do not have Internet access.
- Deny consent to participate.

The criteria for the study interruption were the withdrawal from the patient's part to agree to voluntary participation in the study. The number of subjects needed was estimated to evaluate a difference between groups of 0.5% in HbA1c levels considering a standard deviation of 1.0% (based on the data of available literature). A type I error alpha = 0.05 and a power of 80% were considered in sample size calculation. A sample of 75 subjects for group was needed.

Data were expressed as mean ± standard deviation for continuous variables and percentages for categorical variables. The Kolmogorov–Smirnov test was used to test the normality of distribution of continuous variables. Differences in clinical characteristics were evaluated and compared using the χ^2 tests for categorical variables and the Mann–Whitney test for continuous variables.

Effects of teleconsultation on HbA1c levels were assessed with the use of mixed models [18] that included all observed longitudinal data.

A p value < 0.05 was considered for statistical significance. Analyses were performed using SAS 9.3 (SAS Institute Inc.).

Recruitment occurred during regularly scheduled visits, from December 2014 to January 2016.

Project costs

The Italian Diabetes Foundation—thanks to a financial support received from Intesa Sanpaolo Foundation—covered the overall costs of the project. The overall costs included the specialist consultations, the costs of setting up, and the management of the Web site.

Ethical aspects

All the patients participating in the study signed the consent form for data management in accordance with Legislative Decree no. 196/2003 (protection of personal data). The study was conducted according to the Declaration of Helsinki and subsequent revisions (2008: Sixth and final revision, 59th Meeting, Seoul). All documents related to the study (protocol, informed consent form, etc.) were approved by the Ethics Committee.

Results

The enrollment of patients was stopped at 77 patients: The financial support was stopped due to the prolongation of the activities over the planned timeline. For this reason, a futility analysis was performed. A non-planned futility analysis was performed when the trial was terminated for lack of funding resources. We performed a power calculation based on the between groups found mean HbA1c reduction (0.1%) and its standard deviation (0.7%) with the sample we enrolled (alpha = 0.05). The power was 0.10%, highly lower than that (80%) considered when the pre-trial sample size calculation was performed. We also performed a new sample size calculation based on the real standard deviation of our population (0.7%) and a hypothetic between-groups HbA1c reduction of 0,5% with alpha = 0.05 and power = 80%. With this calculation, a sample of 32 for group was needed.

A total of 77 Caucasian patients affected by type 1 diabetes met the inclusion criteria. Once the informed consent was obtained, they were randomly (1:1) assigned to one of the treatment groups.

40 were randomized to Group A (standard outpatient periodic evaluations).

37 were randomized to Group B (teleconsultation via web).

The dropout rate for the group A was 1 out of 40 patients (2.5%), and for the group B was 2 out of 37 patients (5.4%).

Baseline clinical characteristics of the study population are reported in Table 1.

No statistically significant differences between-group at baseline were identified for patient age, duration of diabetes, gender and type of therapy (CSII Continuous subcutaneous insulin infusion or MDI multiple dose insulin), and HbA1c levels (Table 1).

In the teleconsultation group, there were more patients with diabetic complications, in particular diabetic retinopathy and neuropathy, without reaching a statistical significance.

Table 1 Characteristics of patients enrolled in the project

	Outpatient clinic visit	Teleconsultation visit	*P* value
Number of patients	40	37	
Age (years)	34 ± 8	36 ± 12	0.37
Gender (Male/Female)	14/26	11/26	
Years of diabetes	17 ± 8	20 ± 13	0.27
Basal HbA1c (mmol/mol)	59 ± 11	58 ± 12	0.56
Type of treatment (CSII/MDI)	(19/21)	(16/21)	
Diabetic retinopathy	3	8	0.076
Neuropathy	1	5	0.076
Nephropathy	1	0	0.33
Cardiovascular disease	0	2	0.14

Patients' characterization. Data are mean ± SD. NA = not assessed

Changes in Glycemic Control

Patients did not show a statistical change in HbA1c from baseline at 12th month in both groups (HbA1c = 59 ± 11 vs. 59 ± 12 mmol/l $p = 0.56$ in standard care group and HbA1c = 58 ± 12 vs. 56 ± 10 mmol/l $p = 0.45$ telemedicine group, Fig. 2). The between-group comparison for HbA1c levels difference between baseline and 12 months was not statistically different (ΔHbA1c − 1.9 ± 9.0 vs. − 0.8 ± 7.4 for standard care and telemedicine group, respectively, $p = 0.60$). In addition, we find no statistical significant difference between telemedicine and standard outpatient clinic visits after 4 and 8 month, respectively.

Subgroup analyses For change in HbA1c from baseline to 12th month were carried out for age (Group A: 35 ≤ years $p = 0.58$, > 35 years $p = 0.88$; Group B 35 ≤ years $p = 0.42$, > 35 years p 0.89), baseline HbA1c (Group A ≤ 58 mmol/l $p = 0.72$; > 58 mmol/l $p = 0.29$; Group B ≤ 58 mmol/l $p = 0.72$; > 58 mmol/l p 0.29), type of therapy (Group A CSII p 0.97 or MDI $p = 0.65$; Group B CSII $p = 0.83$ or MDI $p = 0.48$), and duration of diabetes (Group A < 20 years $p = 0.36$ or ≥ 20 years $p = 0.66$, Group B < 20 years $p = 0.49$ or ≥ 20 years $p = 0.74$). No statistically significant differences were observed for any subgroup categories described.

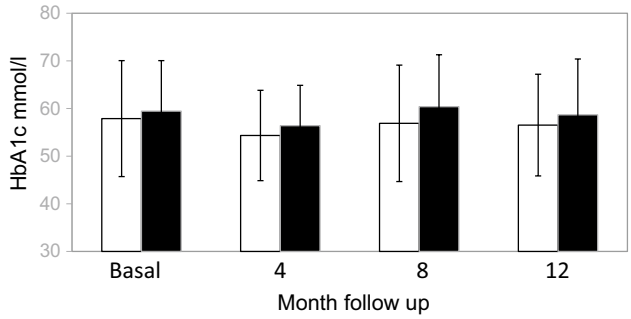

Fig. 2 HbA1c variations during follow-up in patients followed by traditional way (white column) versus patients followed by teleconsultation (black column). HbA1c values at 4, 8, and 12-month follow-up were compared with the corresponding basal value in each group. Variations were not statistically significant. Values are mean ± standard deviation

No variation in the frequency of chronic complications was observed in the two groups. Six emergency room accesses for severe hypoglycemia were reported for two patients of standard visit group (5%). No severe hypoglycemic event was reported in the telemedicine group. The difference was statistically significant ($p = 0.03$). No ketoacidosis episodes were reported in both groups.

In the group A, we registered 171 standard outpatient clinic visits, and in the Group B, 175 telemedicine visits. Eight patients in the telemedicine group needed also additional standard outpatient clinic visits during the study (16 visits overall) due to clinical issues requiring a direct physical examination. The length of the visits has been the same in both groups (A 30 ± 4 min, B 31 ± 3 $p = 0.1$).

Patient satisfactory assessment

At the end of the study, a satisfaction questionnaire was submitted to all the patients followed by teleconsultation. Thirty out of 37 patients completed the survey, of whom 29 out of 30 were fully satisfied of the service (5/5 points) (Table 2). All the patients reported a high level of comfort (5/5 points)

Table 2 Results from the questionnaire of the project evaluation

Topic	Score				
	5	4	3	2	1
Satisfaction	29	1	–	–	–
Advantages					
Improvement of diabetes management	19	3	4	1	3
Time saving	30	–	–	–	–
Cost saving	24	2	1	–	2
Comfort	30	–	–	–	–
Limits of the study					
Unpunctuality	–	–	1	1	28
Technical problems	–	–	1	4	25
Poor interaction with physician	–	–	2	–	28
Score about the services					
Teleconsultation	26	4	–	–	–
Tele-education tools	15	4	5	1	1
Booking system	26	1	–	3	–
Interested to continue					
(Yes/not)	(30/0)				

The score goes to a maximum of 5 to a minimum of 1. Data are number of patients

and 26 out of 30 patients also an improvement of diabetes management (mean of 4.1 points).

Of 30 patients, 28 found no limits of the study. In particular, they did not report as a problem the poor interaction with the physician. The only problem was related to technical problems: Five out of 30 patients found only few and limited technical problems with Internet access. All the patients were interested to continue the project.

Patients' time and cost saving

All the patients of the teleconsultation group reported to have saved (mean time saved for patient 115 ± 86 min for each visit). Furthermore, 24 out of 28 subjects reported a reduction in the costs for the visit (an average cost of 80 euro for each visit, ranging from 10 to 400 euro).

Teleconsultation with nutrition and psychological specialists

Only two of the 37 patients accessed to teleconsultation with psychologist (two and four consultations, respectively) and other four patients with the dietologist.

In the same period, three patients followed in outpatient clinic attended to a psychologist consultation (1, 3, and 2 consultations, respectively) and other three patients to the dietologist.

Discussion

The TELEDIABE main objective was the clinical validation of a new therapeutic strategy aimed to improve the glycemic control and consequently in the long term to reduce the chronic clinical complications in patients affected by type 1 diabetes mellitus under insulin therapy. TELEDIABE proposes a new system for the management of patients with type 1 diabetes mellitus consisting on web-based teleconsultations integrated with some tools for tele-education. Despite several evidences demonstrating the benefits of ICT advanced solution for diabetes management [16, 17, 19–25], the proposal of a teleconsultation with the diabetologists as substitute for the traditional visit in outpatient clinic was never assessed before. Our study is the first structured trial aimed at demonstrating that teleconsultations with video call can be proposed as a real alternative to the standard visit in outpatient clinic.

There are two main concerns in this study. The enrollment of the patients was not concluded due to the economical budget restrictions. Due to some organizational problems, the time for patient enrollment lasted more than expected and we lost the economic support from private foundation. The study was therefore closed ahead of time. This is the reason why the number of enrolled patients was lower than expected. Secondly, the primary outcome, the superiority of teleconsultation versus standard visit, was not reached. The main outcome was probably a bit ambitious. The possibility that the teleconsultation might be superior to our standard visits was probably a difficult challenge: The quality of clinical assistance, the time for the visit, and the physicians involved were the same. A non-inferiority study probably would have been more appropriate. Indeed, according to a post hoc analyses and to the standard variation values of our population, the number of patients enrolled and their relative data might have been sufficient for a non-inferiority study and for proper conclusions.

In addition, our primary outcome (0.5% HbA1c decrease) was defined on the basis of previous publications on population with a poor glycemic control. Indeed, the mean basal HbA1c of our patients indicated an overall metabolic control not far from the target. The consequence is that the primary outcome (a HbA1c decrease of 0.5%) now appears to be a bit ambitious. Therefore, in this population, other expected outcomes might be more adequate to assess the value of our telemedicine approach, i.e., a variation in the incidence of acute complications.

At this regard, no severe hypoglycemic event was reported in the telemedicine group, while six severe hypoglycemic episodes requiring emergency room access were reported in two patients of standard visit group. It is not clear whether the possibility of consulting some educative documents posted online might have been a role in the prevention or treatment of severe acute complications. As showed by the answers to the questionnaire, most of patients reported a good achievement in the management of their diabetes from the project and the treatment of hypoglycemic episodes is one of the topics addressed by the tele-education documents. Notably, the patients followed by teleconsultations had the possibility to book and therefore to access to an additional visit via web easily than the patients followed by standard visit. Therefore, our service of telemedicine probably could reduce hospital admissions as well as specialist and family physician visits, abating the health costs for diabetes. The social impacts will be several: (1) major links among patients and caregivers; (2) quick interface between caregivers and patients affected by diabetes with a gain in quality of life and self-capacity of diabetes management; (3) reduction in necessity of government economical subsidiary intervention for home health and social caregivers to go on site. The acceptability of the new service from the patient's point of view was also measured. All the patients greatly appreciated the service. The degree of acceptability and usability of the technology needs to be validated from the direct users of the innovation. Without the acceptance of the new technology, one cannot expect to achieve any significant results from the development of the technology, both in the experimental

section of the proposal and in the real practice adoption. Patients referred a reduction in the costs related to the diabetes management. A typical cost analysis was not performed, and this potentially impaired the cost estimation. However, these data are reliable for the population we have studied because patients were directly asked to estimate their money and time expenses.

Taken all together these considerations, we believe that our telemedicine service should be considered by the stakeholders. In conclusions, telemedicine for type 1 diabetes mellitus management appeared to have comparable efficacy with standard visits in the overall glucose control but it appeared to reduce the occurrence of severe hypoglycemic episodes. Patients reported high rate of satisfaction, cost and time saved thanks to the service. Furthermore, it ensures a highly customized follow-up, guaranteeing also proximity of the cure to the patients and their families. TELEDIABE represent a real new therapeutic strategy for diabetes management, in future we need more study to verify the feasibility to extend this system also to patients affected by type 2 diabetes.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

Ethical approval This study was approved by the ethics committee (Comitato Etico Milano Area C) (Ethics code: *535-B-102014).

Informed consent Written informed consents were obtained from all participants.

References

1. Farmer A, Gibson OJ, Tarassenko L, Neil A (2005) A systematic review of telemedicine interventions to support blood glucose self-monitoring in diabetes. Diabet Med 22:1372–1378
2. Montori VM, Helgemoe PK, Guyatt GH et al (2004) Telecare for patients with type 1 diabetes and inadequate glycemic control: a randomized controlled trial and meta-analysis. Diabetes Care 27:1088–1094
3. Shulman RM, O'Gorman CS, Palmert MR (2010) The impact of telemedicine interventions involving routine transmission of blood glucose data with clinician feedback on metabolic control in youth with type 1 diabetes: a systematic review and meta-analysis. Int J Pediatr Endocrinol 2010:536957
4. El-Gayar O, Timsina P, Nawar N, Eid W (2013) A systematic review of IT for diabetes self-management: are we there yet? Int J Med Inform 82:637–652
5. Holtz B, Lauckner C (2012) Diabetes management via mobile phones: a systematic review. Telemed J e-Health 18:175–184
6. Guljas R, Ahmed A, Chang K (2014) Whitlock a impact of telemedicine in managing type 1 diabetes among school-age children and adolescents: an integrative review. J Pediatr Nurs 29:198–204
7. Raymond JK, Berget CL, Driscoll K, Ketchum K, Cain C, Thomas F (2016) CoYoT1 clinic: innovative telemedicine care model for young adults with type 1 diabetes. Diabete Technol Ther 18:385–391
8. Marcolino MS, Maia JX, Alkmim MB, Boersma E, Ribeiro AL (2013) Telemedicine application in the care of diabetes patients: systematic review and meta-analysis. PLoS ONE 8:e79246
9. Shea S, Kothari D, Teresi JA et al (2013) Social impact analysis of the effects of a telemedicine intervention to improve diabetes outcomes in an ethnically diverse, medically underserved population: findings from the IDEATel Study. Am J Public Health 103:1888–1894
10. Carter EI, Nunlee-Bland G, Callender C (2011) A patient centric, provider assisted diabetes telehealth self management intervention for urban minorities. Perspesct Health Inf Manage 8:1b
11. Whittemore R, Jaser SS, Faulkner MS, Murphy K, Delamater A, Grey M (2013) TEENCOPE Research Group. Type 1 diabetes eHealth psychoeducation: youth recruitment, participation, and satisfaction. J Med Internet Res 15:e15
12. Kempf K, Altpeter B, Berger J et al (2017) Efficacy of the telemedical lifestyle intervention program TeLiPro in advanced stages of type 2 diabetes: a randomized controlled trial. Diabetes Care 40(7):863–871
13. So CF, Chung J (2017) Telehealth for diabetes self-management in primary healthcare—a systematic review and meta-analysis. J Telemed Telecare 1:1357633X17700552
14. Porter J, Huggins CE, Truby H, Collins J (2016) The effect of using mobile technology-based methods that record food or nutrient intake on diabetes control and nutrition outcomes: a systematic review. Nutrients 8(12):815
15. Viana L, Gomes M, Zajdenverg L, Pavin E, Azevedo M (2016) Brazilian Type 1 Diabetes Study Group interventions to improve patients' compliance with therapies aimed at lowering glycated hemoglobin (HbA1c) in type 1 diabetes: systematic review and meta-analyses of randomized controlled clinical trials of psychological, telecare, and educational interventions. Trials 17:94
16. Toma T, Athanasiou T, Harling L, Darzi A, Ashrafian H (2014) Online social networking services in the management of patients with diabetes mellitus: systematic review and meta-analysis of randomised controlled trials. Diabetes Res Clin Pract 106:200–211
17. Su D, Zhou J, Kelley MS et al (2016) Does telemedicine improve treatment outcomes for diabetes? A meta-analysis of results from 55 randomized controlled trials. Diabetes Res Clin Prac 116:136–148
18. Verbeke G, Molenberghs G (2000) Linear mixed models for longitudinal data. Springer, New York
19. Wojcicki J, Ladyzynski P, Foltynski P (2013) What we can really expect from telemedicine in intensive diabetes treatment: 10 years later. Diabetes Technol Ther 15:3
20. Faruque L, Wiebe N, Ehteshami-Afshar A et al (2017) Alberta kidney disease network effect of telemedicine on glycated hemoglobin in diabetes: a systematic review and meta-analysis of randomized trials. CMAJ 189(9):E341–E364
21. Liu W, Saxon DR, McNair B, Sanagorski R, Rasouli N (2016) Endocrinology telehealth consultation improved glycemic control similar to face-to-face visits in veterans. J Diabetes Sci Technol 10:1079–1086
22. Landau Z, Mazor-Aronovitch K, Boaz M et al (2010) The effectiveness of Internet-based blood glucose monitoring system on improving diabetes control in adolescent with type 1 diabetes. Pediatr Diabetes 13:203–207
23. Rossi MC, Nicolucci A, Lucisano G et al (2013) Impact of the "diabetes interactive diary" telemedicine system on metabolic control, risk of hypoglycemia and quality of life: a randomized trial in type 1 diabetes. Diabetes Technol Ther 15:670–679
24. Pressman AR, Kinoshita L, Kirk S, Barbosa GM, Chou C, Minkoff J (2014) A novel telemonitoring device for improving

diabetes control: protocol and results from a randomized clinical trial. Telemed e-Health 20:109–114
25. Boyle L, Grainger R, Hall RM, Krebs JD (2017) Use of and beliefs about mobile phone apps for diabetes self-management: surveys of people in a hospital diabetes clinic and diabetes health professionals in New Zealand. JMIR Mhealth Uhealth 5(6):e85